

Representation of follow-up

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June 2026

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From C:\Bendix\teach\SPE\git\SPE\lectures\time-rep\time-rep.tex

Friday 29 May, 2026, 07:27

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Representation of follow-up

Bendix Carstensen

Representation of follow-up

SPE, Tartu, Estonia,

June 2026

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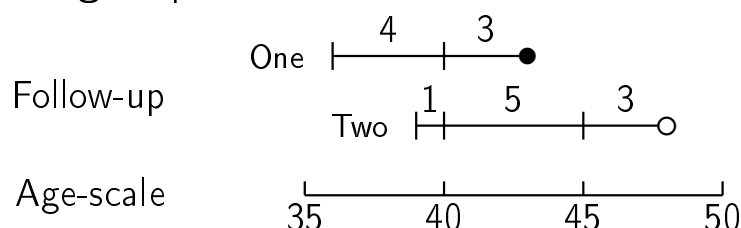
time-split

- ▶ In follow-up studies we estimate rates from:
 - ▶ D — events, deaths
 - ▶ Y — person-years
 - ▶ $\hat{\lambda} = D/Y$ rates
 - ▶ ... empirical counterpart of intensity — an **estimate**
- ▶ Rates differ between persons.
- ▶ Rates differ **within** persons:
 - ▶ by age
 - ▶ by calendar time
 - ▶ by disease duration
 - ▶ ...
- ▶ Multiple timescales.
- ▶ Rates between multiple states (little boxes — later)

Stratification by age

If follow-up is rather short, age at entry is OK for age-stratification.

If follow-up is long, stratification by categories of **current age** is preferable.



— allowing rates to vary across age-bands

— how do we split follow-up by age and why is it OK?

Representation of follow-up (time-split)

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Statistical model for follow-up data

► Data:

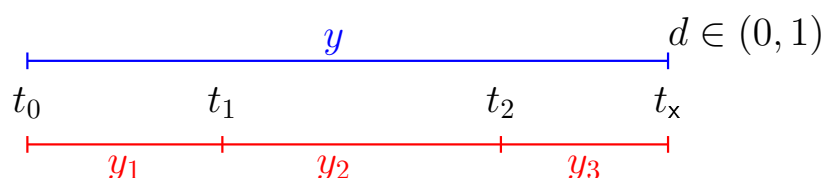
- status and time at entry
- status and time at exit
- ... events (= change of status) and **observed** risk time:
empirical occurrence rates (d, y)

► Model for occurrence rates:

- $\lambda(t, x) = P\{\text{event in } (t, t + dt] | \text{alive at } t, x\} / dt$
- parametric specification of how λ depends on t and x
- likelihood is a function of λ and **data**: $P\{\text{data} | \text{model}, \lambda\}$
- simplest case with constant λ : log-likelihood = $d \log(\lambda) - \lambda y$
- log-likelihood for a Poisson variate d with expectation λy is:
 $d \log(\lambda) - \lambda y$, the same as the rate log-likelihood
- rate model is not a Poisson **model**, but the **likelihood** is the same

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Probability

$P(d \text{ at } t_x | \text{entry } t_0)$

$$\begin{aligned}
 &= P(\text{surv } t_0 \rightarrow t_1 | \text{entry } t_0) \\
 &\times P(\text{surv } t_1 \rightarrow t_2 | \text{entry } t_1) \\
 &\times P(d \text{ at } t_x | \text{entry } t_2)
 \end{aligned}$$

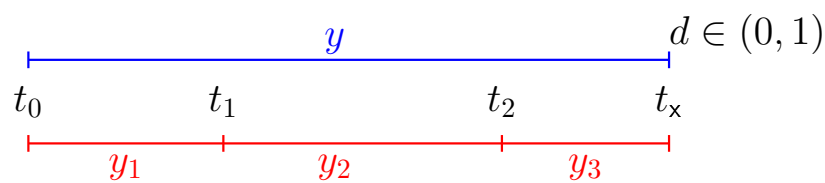
log-Lik (λ constant)

$d \log(\lambda) - \lambda y$

$$\begin{aligned}
 &= 0 \log(\lambda) - \lambda y_1 \\
 &+ 0 \log(\lambda) - \lambda y_2 \\
 &+ d \log(\lambda) - \lambda y_3
 \end{aligned}$$

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Probability

log-Lik (λ varies)

$P(d \text{ at } t_x | \text{entry } t_0)$

$= P(\text{surv } t_0 \rightarrow t_1 | \text{entry } t_0)$

$= 0 \log(\lambda_1) - \lambda_1 y_1$

$\times P(\text{surv } t_1 \rightarrow t_2 | \text{entry } t_1)$

$+ 0 \log(\lambda_2) - \lambda_2 y_2$

$\times P(d \text{ at } t_x | \text{entry } t_2)$

$+ d \log(\lambda_3) - \lambda_3 y_3$

—allows different rates (λ_i) **between** intervals,
but assumes constant rates **within** intervals

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Dividing time in small intervals requires:

time scale / origin: date where the time scale is 0:

- ▶ Age — 0 at date of birth
- ▶ Disease duration — 0 at date of diagnosis

Intervals: How should time be subdivided:

- ▶ 1-year classes? 5-year classes? Equal length?
- ▶ —short enough for assumption of constant rate in each interval

Aim: Separate rate in each interval

—mimicking continuous time using short intervals with **constant rate**
—time at the beginning of interval used as **quantitative variable**.

Representation of follow-up (time-split)

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Follow-up intervals on several timescales

- ▶ The risk-time is the same on all timescales
- ▶ So only need the **entry** point on each time scale:
 - ▶ Age at entry
 - ▶ Date of entry
 - ▶ Time since diagnosis at entry
 - if time of diagnosis is the entry, this is 0 for all.

- ▶ **Response variable** in analysis of rates:

(d, y) (**event, duration**)

- ▶ **Covariates** in analysis of rates:

- ▶ **timescales**
- ▶ other (fixed) measurements

- ▶ ... do not confuse **duration** and **timescale** !

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Follow-up data in Epi — Lexis objects I

```
> thoro[1:4,1:8]

  id sex birthdat contrast injecdat volume  exitdat exitstat
1  1  2  1916.609          1  1938.791     22  1976.787         1
2  2  2  1927.843          1  1943.906     80  1966.030         1
3  3  1  1902.778          1  1935.629     10  1959.719         1
4  4  1  1918.359          1  1936.396     10  1977.307         1

> thL <- Lexis(entry = list(age = injecdat - birthdat,
+                             dat = injecdat,
+                             tfi = 0),
+             exit = list(dat = exitdat),
+             exit.status = factor(exitstat == 1,
+                                   labels = c("Alive", "Dead")),
+             data = thoro)

NOTE: entry.status has been set to "Alive" for all.
NOTE: Dropping 2 rows with duration of follow up < tol
```

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Follow-up data in Epi — Lexis objects II

```
> summary(thL, timeScales = TRUE)

Transitions:
  To
From  Alive Dead Records: Events: Risk time: Persons:
  Alive  504 1964    2468    1964   51934.08    2468

Timescales:
age dat tfi
""  ""  ""
```

Representation of follow-up (time-split)

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Definition of Lexis object

```
thL <- Lexis(entry = list(age = injecdat - birthdat,
                           dat = injecdat,
                           tfi = 0),
             exit = list(dat = exitdat),
             exit.status = factor(exitstat == 1,
                                   labels = c("Alive","Dead")),
             data = thoro)
```

entry is defined on **three** timescales,
but **exit** is only needed on **one** timescale (or vice versa):
Follow-up time is the same on all timescales: **exitdat - injecdat**
One element of **entry** and **exit** must have same name (here, **dat**).

Representation of follow-up (time-split)

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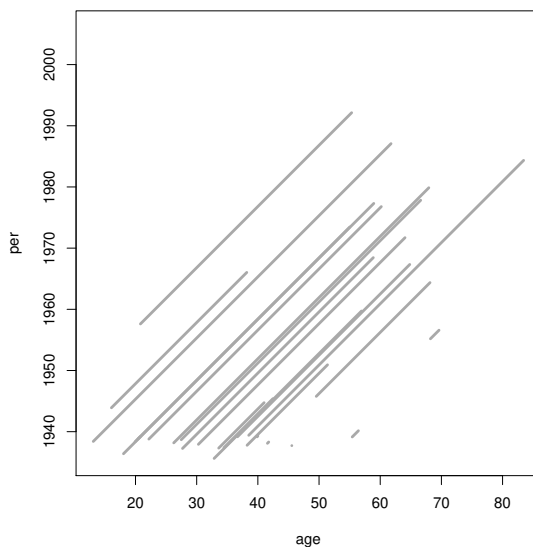
The looks of a Lexis object

```
> thL[1:4,1:9]
  age    dat tfi lex.dur lex.Cst lex.Xst lex.id
1 22.18 1938.79  0  37.99  Alive  Dead    1
2 49.54 1945.77  0  18.59  Alive  Dead    2
3 68.20 1955.18  0   1.40  Alive  Dead    3
4 20.80 1957.61  0  34.52  Alive  Alive    4
...

> summary(thL)
Transitions:
  To
From Alive Dead Records: Events: Risk time: Persons:
  0   504 1964      2468     1964    51934.08     2468
```

Representation of follow-up (time-split)

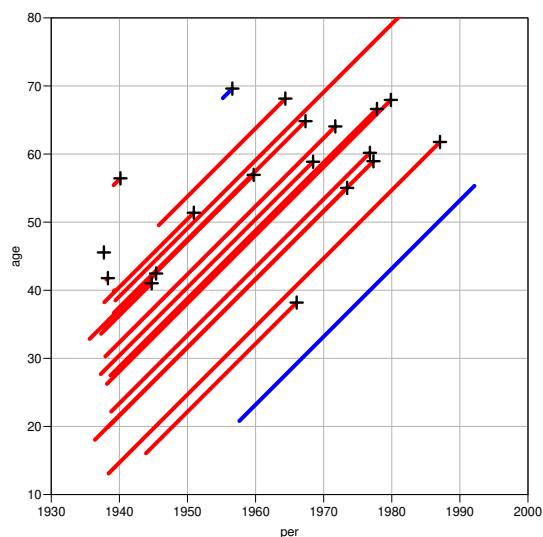
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```
> plot( thL, lwd=3 )
```

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Lexis diagram

```
> plot(thL, 2:1, lwd=5, col=c("red", "blue")[thL$contrast],
+       grid = TRUE, lty.grid = 1, col.grid = gray(0.7),
+       xlim = 1930 + c(0,70), xaxs = "i", ylim = 10 + c(0, 70), yaxs = "i", las = 1 )
> points( thL, 2:1, pch=c(NA,3)[thL$lex.Xst], lwd = 3, cex = 1.5 )
```

Representation of follow-up (time-split)

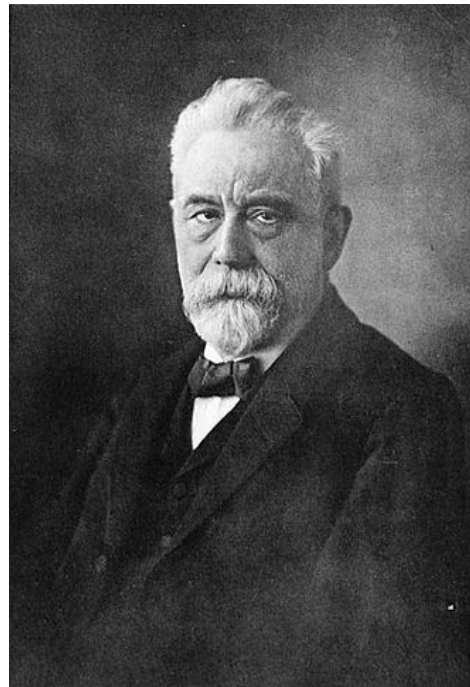
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EINLEITUNG IN DIE THEORIE DER BEVÖLKERUNGSSTATISTIK

VON
W. LEXIS
DR. DER STAATSWISSENSCHAFTEN UND DER PHILOSOPHIE,
O. PROFESSOR DER STATISTIK IN DORPAT.

STRASSBURG
KARL J. TRUBNER

Representation of follow-up (time-split) 1875.



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Splitting follow-up time

```
> spl1 <- splitLexis(thL, time.scale = "age", breaks = seq(0, 100, 20))
> round(spl1, 1)
  age    dat   tfi lex.dur lex.Cst lex.Xst lex.id sex birthdat contrast injecdat vo
22.2 1938.8  0.0   17.8      0      0      1  2   1916.6          1   1938.8
40.0 1956.6 17.8   20.0      0      0      1  2   1916.6          1   1938.8
60.0 1976.6 37.8    0.2      0      1      1  2   1916.6          1   1938.8
49.5 1945.8  0.0   10.5      0      0     640  2   1896.2          1   1945.8
60.0 1956.2 10.5    8.1      0      1     640  2   1896.2          1   1945.8
68.2 1955.2  0.0    1.4      0      1    3425  1   1887.0          2   1955.2
20.8 1957.6  0.0   19.2      0      0    4017  2   1936.8          2   1957.6
40.0 1976.8 19.2   15.3      0      0    4017  2   1936.8          2   1957.6
...
```

Representation of follow-up (time-split)

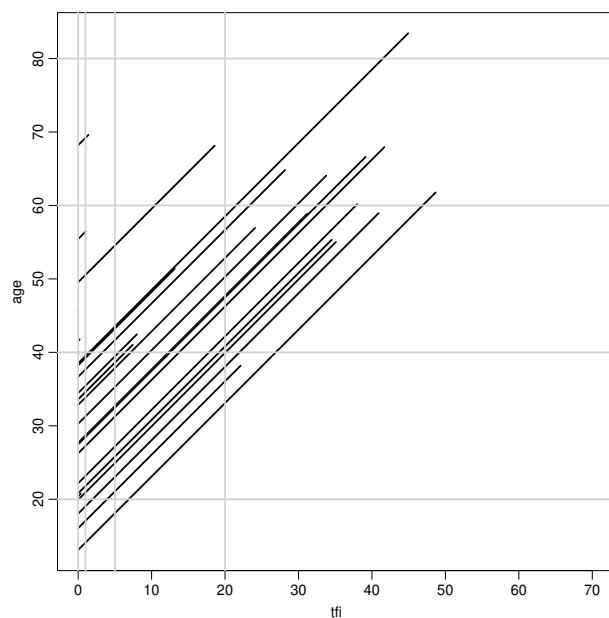
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Split on another timescale

```
> spl2 <- splitLexis(spl1, time.scale="tfi", breaks=c(0,1,5,20,100) )
> round(spl2, 1)
  lex.id age    dat   tfi lex.dur lex.Cst lex.Xst  id sex birthdat contrast inj
1      1 22.2 1938.8  0.0    1.0      0      0   1  2   1916.6          1  1
2      1 23.2 1939.8  1.0    4.0      0      0   1  2   1916.6          1  1
3      1 27.2 1943.8  5.0   12.8      0      0   1  2   1916.6          1  1
4      1 40.0 1956.6 17.8    2.2      0      0   1  2   1916.6          1  1
5      1 42.2 1958.8 20.0   17.8      0      0   1  2   1916.6          1  1
6      1 60.0 1976.6 37.8    0.2      0      1   1  2   1916.6          1  1
7      2 49.5 1945.8  0.0    1.0      0      0  640  2   1896.2          1  1
8      2 50.5 1946.8  1.0    4.0      0      0  640  2   1896.2          1  1
9      2 54.5 1950.8  5.0    5.5      0      0  640  2   1896.2          1  1
10     2 60.0 1956.2 10.5    8.1      0      1  640  2   1896.2          1  1
11     3 68.2 1955.2  0.0    1.0      0      0 3425  1   1887.0          2  1
12     3 69.2 1956.2  1.0    0.4      0      1 3425  1   1887.0          2  1
13     4 20.8 1957.6  0.0    1.0      0      0 4017  2   1936.8          2  1
14     4 21.8 1958.6  1.0    4.0      0      0 4017  2   1936.8          2  1
15     4 25.8 1962.6  5.0   14.2      0      0 4017  2   1936.8          2  1
16     4 40.0 1976.8 19.2    0.8      0      0 4017  2   1936.8          2  1
17     4 40.8 1977.6 20.0   14.5      0      0 4017  2   1936.8          2  1
```

Representation of follow-up (time-split)

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age	tfi	lex.dur	lex.Cst	lex.Xst
22.2	0.0	1.0	0	0
23.2	1.0	4.0	0	0
27.2	5.0	12.8	0	0
40.0	17.8	2.2	0	0
42.2	20.0	17.8	0	0
60.0	37.8	0.2	0	1

plot(spl2, c(1, 3), col = "black", lwd = 2)

Representation of follow-up (time-split)

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Splitting on several timescales

```
> spl1 <- splitLexis(thL, time.scale = "age", breaks = seq(0, 100, 20))
> spl2 <- splitLexis(spl1, time.scale = "tfi", breaks = c(0, 1, 5, 20, 100))
> summary(spl2)
```

Transitions:

To

From Alive Dead Records: Events: Risk time: Persons:
 Alive 8250 1964 10214 1964 51934.08 2468

```
> library(popEpi)
```

```
> splx <- splitMulti(thL, age = seq(0, 100, 20),
+                   tfi = c(0, 1, 5, 20, 100))
```

```
> summary(splx)
```

Transitions:

To

From Alive Dead Records: Events: Risk time: Persons:
 Alive 8248 1964 10212 1964 51916.98 2468

```
> # NOTE: splitMulti excludes follow-up outside range of breaks
```

Representation of follow-up (time-split)

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Likelihood for time-split data

- ▶ We assume that rates are constant in each (small) interval
- ▶ Each observation in the dataset represents an interval, contributing a term to the (log-)likelihood for the rate
- ▶ Each **term** looks like a contribution from a Poisson variate (albeit with values only 0 or 1)
- ▶ So the likelihood from a single **person** looks like the likelihood from several independent Poisson variates
- ▶ ... but the data are neither independent nor Poisson

Representation of follow-up (time-split)

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Analysis of time-split data

Observations (records) classified by p —person and i —interval

- ▶ d_{pi} — events: `lex.Xst=="Dead" & lex.Xst!=lex.Cst`
- ▶ y_{pi} — risk time: `lex.dur` (duration)
- ▶ Covariates are:
 - ▶ timescales (age, period, time since entry)
 - ▶ other variables for this person (constant in each interval).
- ▶ Rate likelihood for **one person** is identical to a Poisson likelihood for **many independent** Poisson variates
- ▶ Modeling rates using `glm` or `gam`:
time-scales are treated as other covariates

Representation of follow-up (time-split)

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Fitting a simple model—data:

```
> stat.table(contrast,
+           list(D = sum(lex.Xst == "Dead"),
+                Y = sum(lex.dur),
+                Rate = ratio(lex.Xst == "Dead", lex.dur, 100)),
+           margin = TRUE,
+           data = spl2)
```

contrast	D	Y	Rate
1	928.00	20094.74	4.62
2	1036.00	31839.35	3.25
Total	1964.00	51934.08	3.78

Representation of follow-up (time-split)

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Fitting a simple model with `poisson`

contrast	D	Y	Rate
1	928.00	20094.74	4.62
2	1036.00	31839.35	3.25

```
> m0 <- glm((lex.Xst == "Dead") ~ factor(contrast) - 1,
+           offset = log(lex.dur / 100),
+           family = poisson,
+           data = spl2)
> round(ci.exp(m0), 2)
```

	exp(Est.)	2.5%	97.5%
factor(contrast)1	4.62	4.33	4.93
factor(contrast)2	3.25	3.06	3.46

... a Poisson model for mortality using log-person-years as offset

Representation of follow-up (time-split)

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Fitting a simple model with `poisreg`

contrast	D	Y	Rate
1	928.00	20094.74	4.62
2	1036.00	31839.35	3.25

```
> m0 <- glm(cbind(lex.Xst == "Dead", lex.dur / 100) ~ factor(contrast) - 1,
+          family = poisreg,
+          data = spl2)
> round(ci.exp(m0), 2)

              exp(Est.) 2.5% 97.5%
factor(contrast)1      4.62 4.33  4.93
factor(contrast)2      3.25 3.06  3.46
```

... a Poisson model for mortality rates based on deaths and person-years

Representation of follow-up (time-split)

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Fitting a simple model with `glmLexis`

The wrapper `glmLexis` requires that `lex.Cst` and `lex.Xst` are factors — see `factorize`:

```
> m0 <- glmLexis(spl2, ~ factor(contrast) - 1, scale = 100)
stats::glm Poisson analysis of Lexis object spl2 with log link:
Rates for the transition:
Alive->Dead
, lex.dur (person-time) scaled by 100
> round(ci.exp(m0), 2)

              exp(Est.) 2.5% 97.5%
factor(contrast)1      4.62 4.33  4.93
factor(contrast)2      3.25 3.06  3.46
```

... a Poisson model for mortality rates based on deaths and person-years in a `Lexis` object

Representation of follow-up (time-split)

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Fitting a simple model — aggregate data

contrast	D	Y	Rate
1	928.00	20094.74	4.62
2	1036.00	31839.35	3.25

As long as we only use covariates that take only a few values, we can model the aggregate data directly:

```
> mx <- glm(cbind(c( 928 , 1036 ),
+                  c(20094.74, 31839.35) / 100) ~ factor(1:2) - 1,
+          family = poisreg )
> round(ci.exp(mx), 2)

              exp(Est.) 2.5% 97.5%
factor(1:2)1      4.62 4.33  4.93
factor(1:2)2      3.25 3.06  3.46
```

Representation of follow-up (time-split)

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SMR

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Representation of follow-up

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SMR

Cohorts where all are exposed

When there is no comparison group we may ask:

Do mortality rates in cohort differ from those of an **external** population, for example:

Rates from:

- ▶ Occupational cohorts
- ▶ Patient cohorts

compared with reference rates obtained from:

- ▶ Population statistics (mortality rates)
- ▶ Hospital registers (disease rates)

Cohort rates vs. population rates: SMR

- ▶ **Multiplicative:** $\lambda(a) = \theta \times \lambda_p(a)$
- ▶ Cohort rates proportional to reference rates, λ_p :
 $\lambda(a) = \theta \times \lambda_p(a)$ — θ the same in all age-bands.
- ▶ D_a deaths during Y_a person-years in age-band a gives the likelihood:

$$\begin{aligned} D_a \log(\lambda(a)) - \lambda(a) Y_a &= D_a \log(\theta \lambda_p(a)) - \theta \lambda_p(a) Y_a \\ &= D_a \log(\theta) + D_a \log(\lambda_p(a)) - \theta (\lambda_p(a) Y_a) \end{aligned}$$

- ▶ The constant $D_a \log(\lambda_p(a))$ does not involve θ , and so can be dropped.

- ▶ $\lambda_p(a)Y_a = E_a$ is the “expected” number of cases in age a , so the log-likelihood contribution from age a is:

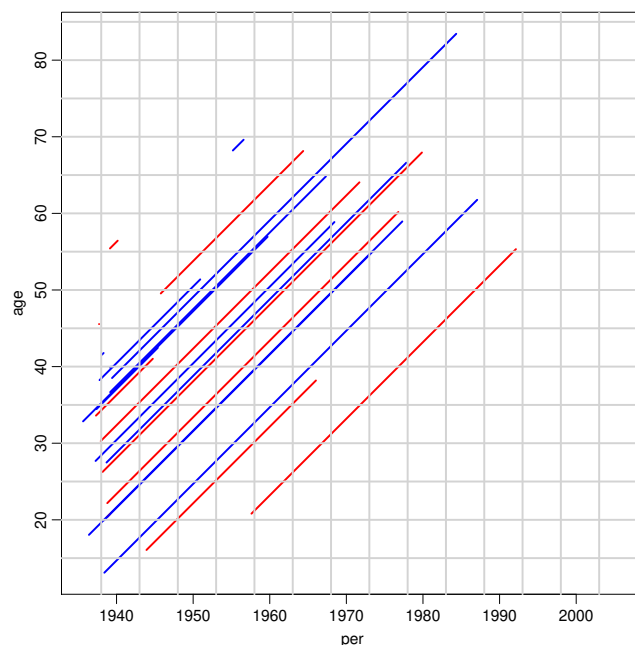
$$D_a \log(\theta) - \theta(\lambda_p(a)Y_a) = D_a \log(\theta) - \theta(E_a)$$

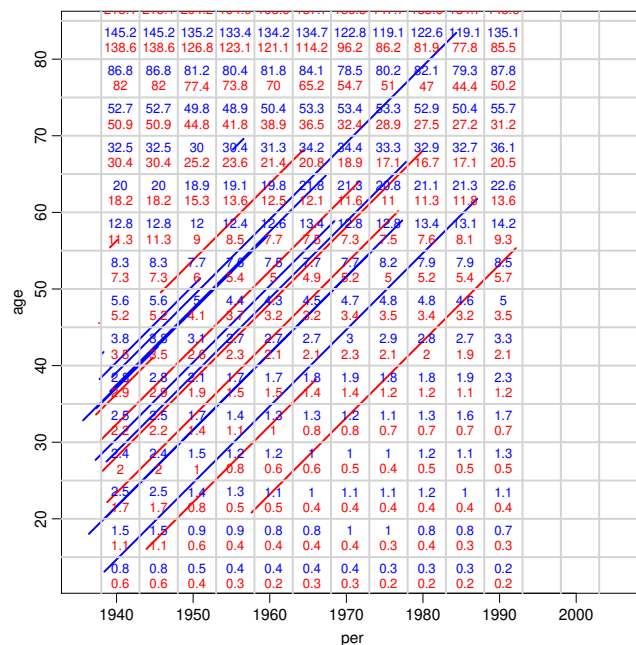
- ▶ The log-likelihood is similar to the log-likelihood for a rate, just with Y replaced by E , so:

$$\hat{\theta} = \sum_a D_a / \sum_a E_a = \text{Observed/Expected} = \text{SMR}$$

Modeling the SMR in practice

- ▶ As for the rates, the SMR can be modelled using interval data.
- ▶ First column of response is d_i , the event indicator (`lex.Xst`).
- ▶ Second column the expected value for each piece of follow-up, $e_i = y_i \times \lambda_p$ (`lex.dur * rate`)
- ▶ λ_p is the population rate corresponding to the age, period and sex of the follow-up period y_i .





SMR (SMR)

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Split the data to fit with population data

```
> thad <- splitMulti(thL, age=seq(0,90,5), dte=seq(1938,2038,5) )
> summary( thad )
```

Transitions:

To

```
From 0 1 Records: Events: Risk time: Persons:
0 21059 1939 22998 1939 51787.96 2463
```

Create variables to fit with the population data

```
> thad$agr <- timeBand( thad, "age", "left" )
> thad$per <- timeBand( thad, "dte", "left" )
> round( thad[1:5,c("lex.id","age","agr","dte","per","lex.dur","lex.Xst","sex")],

lex.id age dte lex.dur lex.Xst agr per sex
1 22.18 1938.79 2.82 0 20 1938 2
1 25.00 1941.61 1.39 0 25 1938 2
1 26.39 1943.00 3.61 0 25 1943 2
1 30.00 1946.61 1.39 0 30 1943 2
1 31.39 1948.00 3.61 0 30 1948 2
```

SMR (SMR)

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```
> data( gmortDK )
> dim( gmortDK )
[1] 418 21
> gmortDK[1:6,1:6]

agr per sex risk dt rt
1 0 38 1 996019 14079 14.135
2 5 38 1 802334 726 0.905
3 10 38 1 753017 600 0.797
4 15 38 1 773393 1167 1.509
5 20 38 1 813882 2031 2.495
6 25 38 1 789990 1862 2.357

> gmortDK$per <- gmortDK$per+1900
> thadx <- merge( thad, gmortDK[,c("agr","per","sex","rt")] )
> thadx$E <- thadx$lex.dur * thadx$rt / 1000
```

SMR (SMR)

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contrast	D	Y	E	SMR
1	917.00	20045.46	214.66	4.27
2	1022.00	31742.51	447.21	2.29

```
> m.SMR <- glm(cbind(lex.Xst, E) ~ factor(contrast) - 1,
+             family = poisreg,
+             data = thadx)
> round(ci.exp(m.SMR), 2)
```

```
               exp(Est.) 2.5% 97.5%
factor(contrast)1      4.27 4.00  4.56
factor(contrast)2      2.29 2.15  2.43
```

- Analysis of SMR is like analysis of rates:
- Replace Y with E — that's all! (`glmLexis` not usable)
- ...it's the calculation of E that is difficult

SMR (SMR)

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RSR

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RSR

Cohort rates vs. population rates: RSR

- **Additive:** $\lambda(a) = \delta(a) + \lambda_p(a)$, λ_p assumed known
- Note that the survival (since $a = a_0$, say) is:

$$\begin{aligned}
 S(a) &= \exp\left(-\int_{a_0}^a \delta(a) + \lambda_p(a) da\right) \\
 &= \exp\left(-\int_{a_0}^a \delta(a) da\right) \times \exp\left(-\int_{a_0}^a \lambda_p(a) da\right) \\
 \Rightarrow r(a) &= S(a)/S_p(a) = \exp\left(-\int_{a_0}^a \delta(a) da\right)
 \end{aligned}$$

- **Additive** model for **rates** $\Leftrightarrow$ **Relative survival** model.

RSR (RSR)

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Split the data to fit with population data

```
> data( thoro )
> thoro <- cal.yr(thoro)
> thL <- Lexis(entry = list(age = injecdat - birthdat,
+                           dte = injecdat,
+                           tfi = 0),
+             exit = list(dte = exitdat),
+             exit.status = as.numeric(exitstat == 1),
+             data = thoro)

NOTE: entry.status has been set to 0 for all.
NOTE: Dropping 2 rows with duration of follow up < tol

> thad <- splitMulti(thL, age = seq(0, 90, 5), dte = seq(1938, 2038, 5))
> thad$agr <- timeBand(thad, "age", "left")
> thad$per <- timeBand(thad, "dte", "left")
> thad[1:5, c(lexVars(thad), "agr", "per", "sex")]
```

	lex.id	age	dte	tfi	lex.dur	lex.Cst	lex.Xst	agr	per	sex
	1	22.18	1938.79	0.00	2.82	0	0	20	1938	2
	1	25.00	1941.61	2.82	1.39	0	0	25	1938	2
	1	26.39	1943.00	4.21	3.61	0	0	25	1943	2
	1	30.00	1946.61	7.82	1.39	0	0	30	1943	2
RSR (RSR)	1	31.39	1948.00	9.21	3.61	0	0	30	1948	2

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Important to have rates in correct dimension

```
> data(gmortDK)
> gmortDK[1:3, 1:6]

  agr per sex  risk    dt    rt
1   0  38   1 996019 14079 14.135
2   5  38   1 802334   726  0.905
3  10  38   1 753017   600  0.797

> gmortDK <- mutate(gmortDK, per = per + 1900,
+                   rt = rt / 1000) # rates per 1 year
> thadx <- merge( thad, gmortDK[,c("agr","per","sex","rt")] )
> thadx <- mutate( thadx, E = lex.dur * rt )
```

RSR (RSR)

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Additive excess rate model—data

```
> (stat.table( contrast,
+             list(D = sum(lex.Xst),
+                   Y = sum(lex.dur),
+                   Rate = ratio(lex.Xst, lex.dur, 100),
+                   E = sum(E),
+                   Erate = ratio(E, lex.dur, 100),
+                   SMR = ratio(lex.Xst, E)),
+             data = thadx ) -> zz)
```

contrast	D	Y	Rate	E	Erate	SMR
1	917.00	20045.46	4.57	214.66	1.07	4.27
2	1022.00	31742.51	3.22	447.21	1.41	2.29

```
> round(cbind(cohR = zz["Rate", ], popR = zz["Erate", ],
+             excR = zz["Rate", ] - zz["Erate", ]), 2)
```

	cohR	popR	excR
1	4.57	1.07	3.50
2	3.22	1.41	1.81

RSR (RSR)

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Simple additive model

A model where the **difference** between the cohort and the population rates are the parameters.

Poisson model with **identity link** and the expected numbers as **offset**

... not easy to fit:

```
> try(
+ glm(cbind(lex.Xst == 1, lex.dur) ~ factor(contrast) - 1,
+     offset = E,
+     family = poisreg(link = "identity"),
+     data = thadx) ); geterrmessage(); warnings()

[1] "Error : no valid set of coefficients has been found: please supply starting
Advarselsbesked:
In log(y/mu) : NaNs produced
```

RSR (RSR)

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Additive excess rate model

```
> # start without offset
> glm(cbind(lex.Xst == 1, lex.dur) ~ factor(contrast) - 1,
+     family = poisreg(link = "identity"),
+     data = thadx) -> m0
> #
> # add offset and use starting values from simple model
> update(m0, offset = E,
+       start = coef(m0)) -> mE
> round(ci.lin(mE), 4)

              Estimate StdErr          z P    2.5%  97.5%
factor(contrast)1    0.0310 0.0015 20.0661 0 0.0280 0.0340
factor(contrast)2    0.0179 0.0010 17.1888 0 0.0158 0.0199

> round(ci.exp(mE, Exp = FALSE) * 100, 1) # excess mortality in %/year

              Estimate 2.5% 97.5%
factor(contrast)1      3.1  2.8  3.4
factor(contrast)2      1.8  1.6  2.0
```

RSR (RSR)

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Additive excess rate model—estimates

```
> round(ci.exp(mE, Exp = FALSE) * 100, 2) # excess mortality in %/year

              Estimate 2.5% 97.5%
factor(contrast)1      3.10 2.80  3.40
factor(contrast)2      1.79 1.58  1.99

> round(cbind(cohR = zz["Rate", ],
+             popR = zz["Erate", ],
+             excR = zz["Rate", ] - zz["Erate", ]), 2)

      cohR popR excR
1 4.57 1.07 3.50
2 3.22 1.41 1.81
```

Not identical estimates, because the model is with **identity link**—not the canonical link for the Poisson distribution.

For **contrast=1** the excess rate is 3.1 per 100 PY

For **contrast=2** the excess rate is 1.8 per 100 PY

RSR (RSR)

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